

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
PUDUCHERRY TECHNOLOGICAL UNIVERSITY**

**MCA -FINAL YEAR PROJECT(IV-SEM)
FIRST REVIEW**

**TITLE
REAL ESTATE BUSINESS MANAGEMENT
SYSTEM**

Project Guide
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Submitted By:-
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1) Modules Descriptions:

1. User Management Module:

This module handles user registration, login, and role-based access control for buyers, sellers, and agents (admins). Buyers can create accounts to browse and book properties, while sellers can register to list their properties for sale. Agents (admins) oversee the system, verifying user details and managing platform security.

2. Property Management Module:

Sellers can list, update, and remove properties, while admins verify and approve listings. Each property includes details such as price, location, size, and images. This module ensures that only verified listings are displayed to buyers, enhancing the credibility of the platform.

3. Property Search & View Module

Buyers can browse and search for properties using filters like price, location, and property type. The system fetches and displays property details, including high-quality images, descriptions, and agent contact information. This module helps buyers make informed decisions before proceeding with bookings.

4. Booking & Appointment Module

This module allows buyers to schedule property visits and make booking requests. Sellers and agents receive notifications and can accept or reject appointment requests. It ensures smooth communication between all parties, enabling efficient scheduling and tracking of bookings.

5. Payment & Transaction Module

Buyers can make secure online payments to confirm property bookings. Payment details are recorded, and receipts are generated for transparency. Admins monitor all transactions, ensuring security and compliance with financial regulations.

6. Booking Confirmation Module:

Once payment is successful, the system confirms the booking and sends notifications to buyers, sellers, and agents. Confirmation details are stored, and users receive updates via messages. This module ensures timely communication and transparency in transactions.

7. Admin Dashboard Module

Agents (admins) oversee platform operations, managing users, properties, and transactions. They can approve property listings, resolve disputes, and track booking trends. This module provides a centralized interface for monitoring and controlling all activities within the system.

8. Reports & Analytics Module

The system generates reports on property bookings, transactions, and user activity. Admins can analyse sales trends, track revenue, and identify popular property locations. This module provides valuable insights to enhance platform performance and decision-making.

2) Design:

a. Detailed Class Diagram:

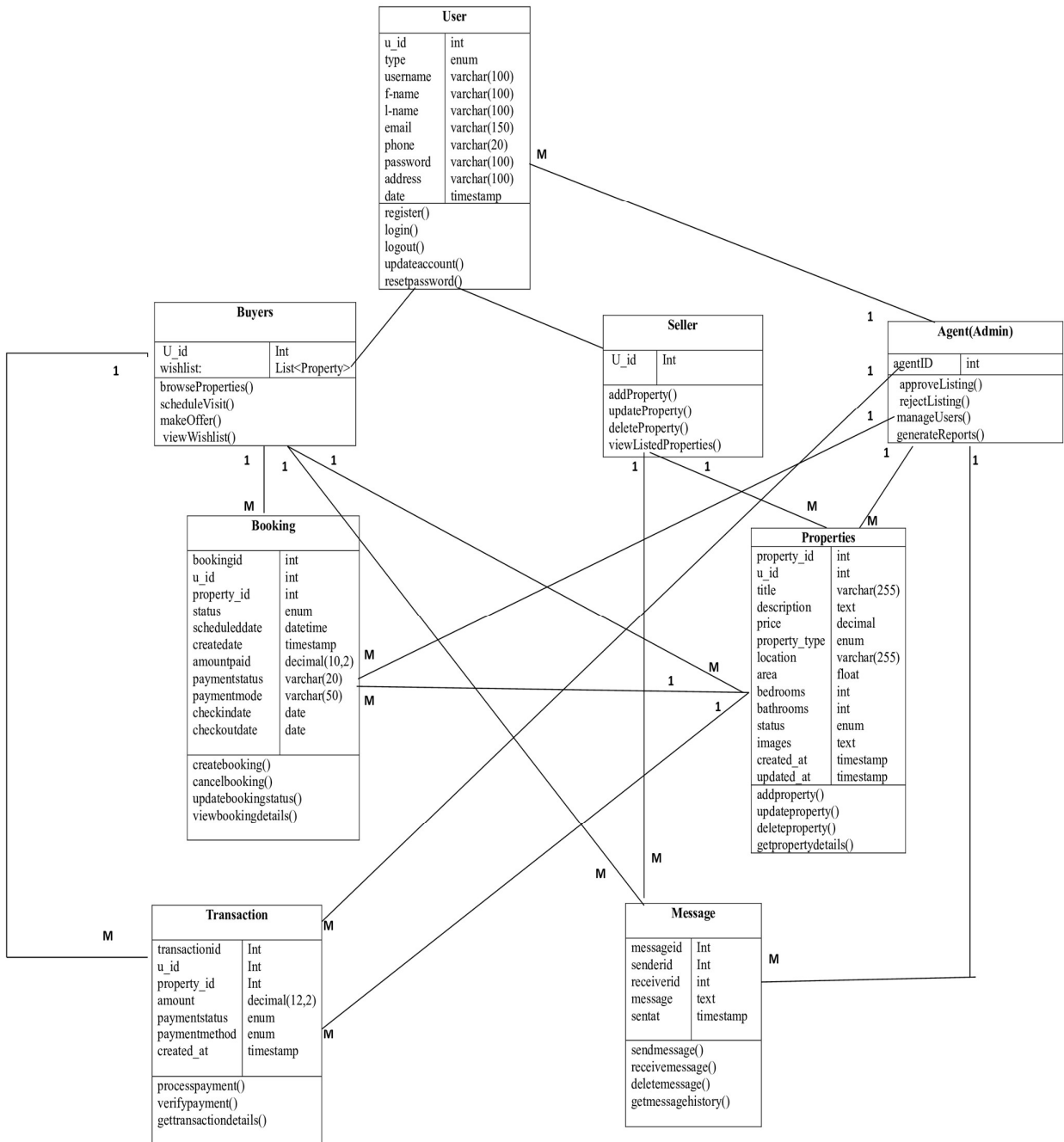


Fig 2(a): Class Diagram

b. Sequence Diagram:

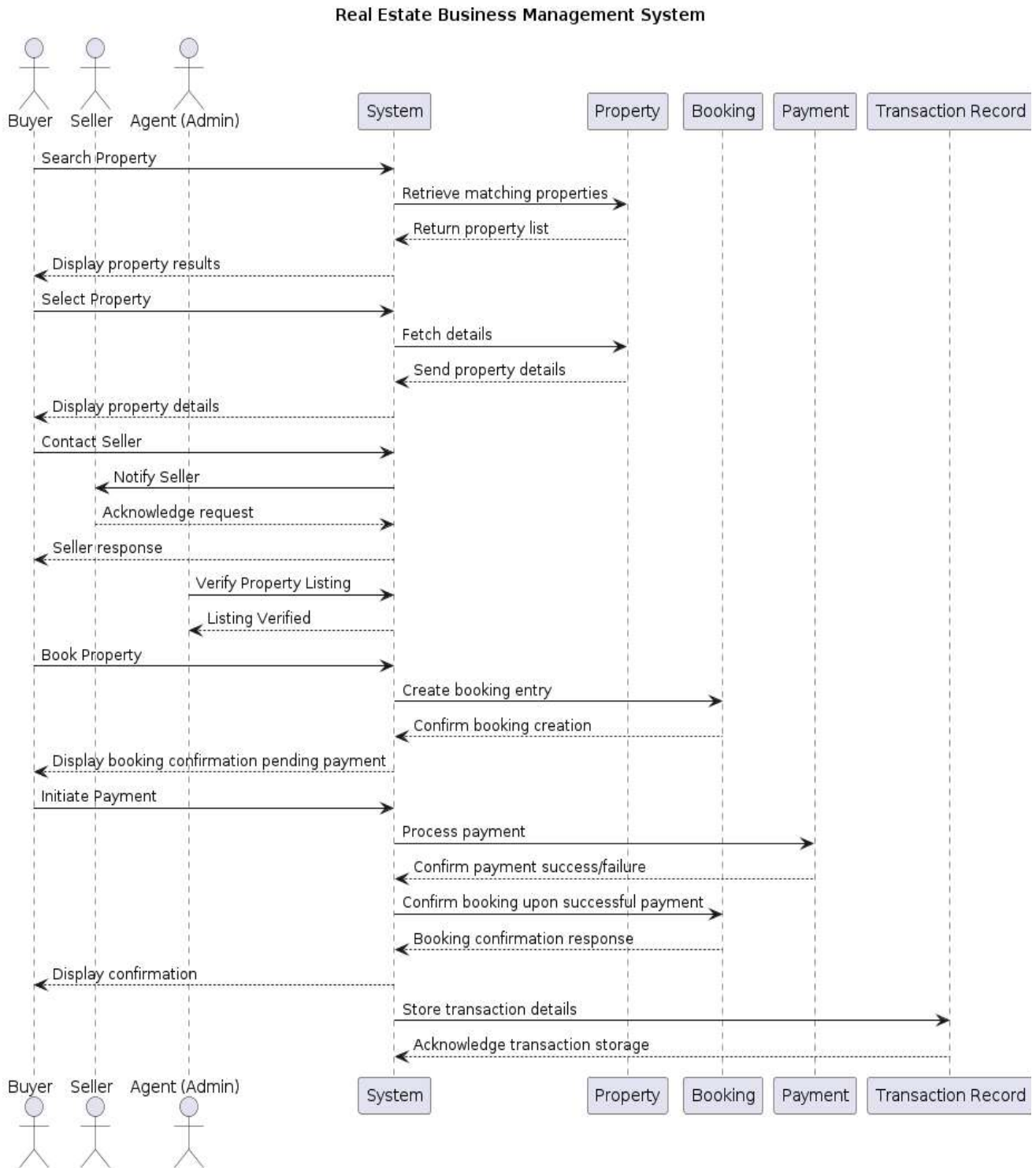


Fig 2(b): Sequence Diagram

c. State Chart Diagram:

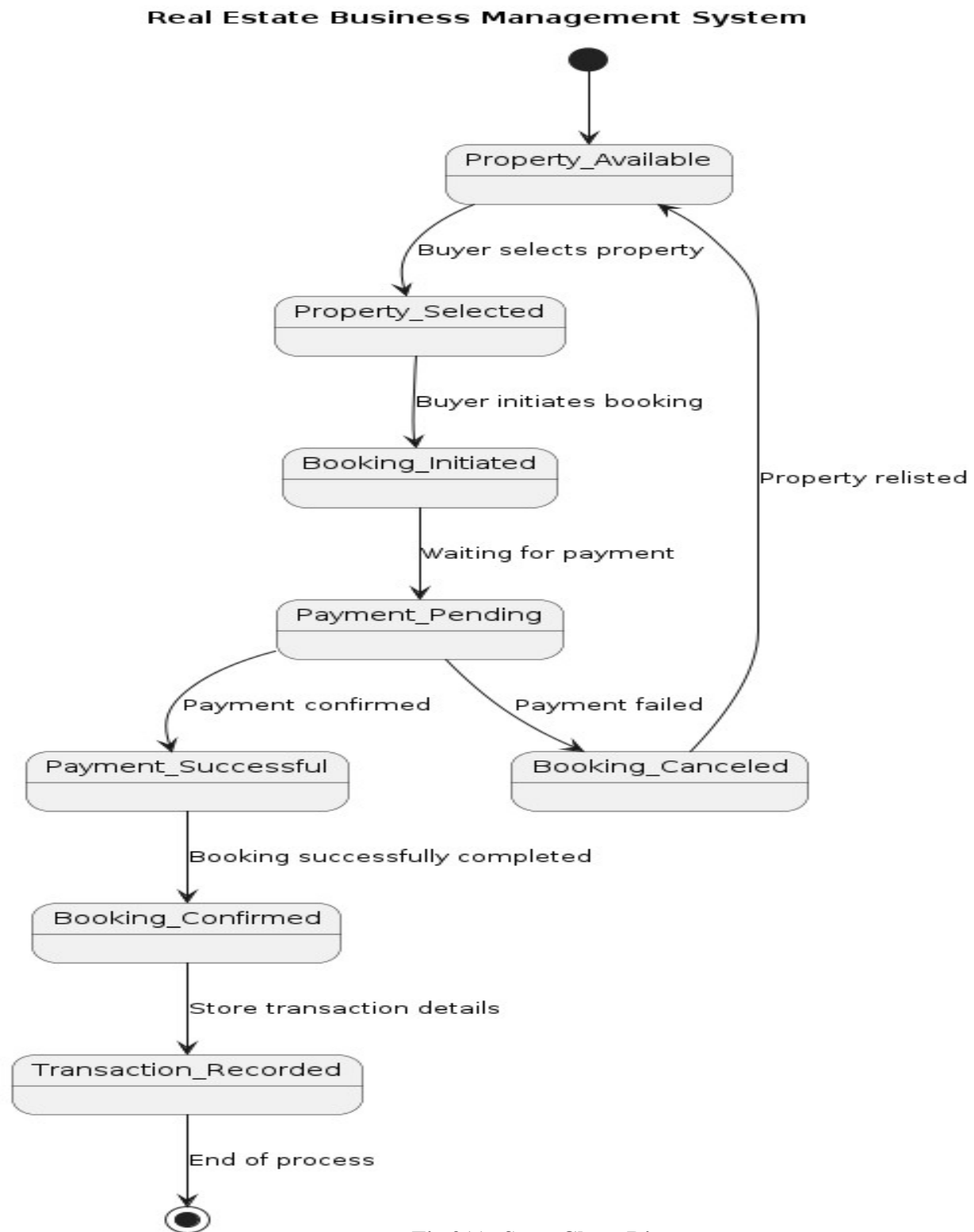


Fig 2(c): State Chart Diagram

d. Activity Diagram:

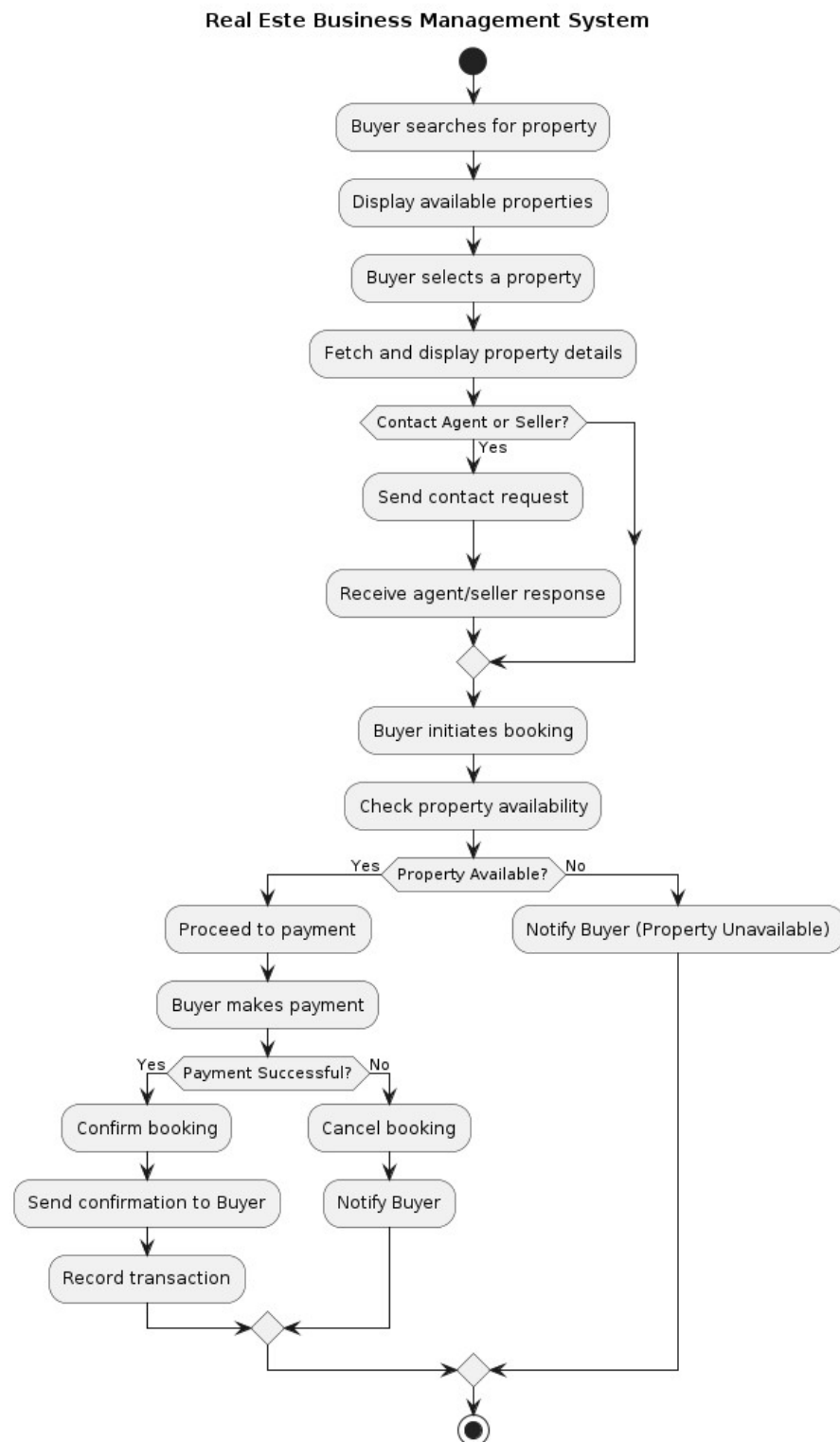


Fig 2(d): Activity Diagram

3) Data Base Design:

a. Users:

Field	Type	Null	Key	Default	Extra
u_id	int(11)	NO	PRI	NULL	auto_increment
type	enum('buyer','seller')	NO		NULL	
username	varchar(100)	NO	UNI	NULL	
f_name	varchar(100)	NO		NULL	
l_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	
phone	varchar(20)	NO	UNI	NULL	
password	varchar(255)	NO		NULL	
address	text	NO		NULL	
date	timestamp	NO		CURRENT_TIMESTAMP	on update CURRENT_TIMESTAMP

b. Properties:

Field	Type	Null	Key	Default	Extra
property_id	int(11)	NO	PRI	NULL	auto_increment
u_id	int(11)	NO	MUL	NULL	
title	varchar(255)	NO		NULL	
description	text	NO		NULL	
price	decimal(12,2)	NO		NULL	
property_type	enum('house','apartment','land','commercial')	NO		NULL	
location	varchar(255)	NO		NULL	
area	float	NO		NULL	
bedrooms	int(11)	YES		0	
bathrooms	int(11)	YES		0	
status	enum('available','sold','rented')	YES		available	
images	text	YES		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	
updated_at	timestamp	NO		CURRENT_TIMESTAMP	on update CURRENT_TIMESTAMP

c. Bookings:

Field	Type	Null	Key	Default	Extra
booking_id	int(11)	NO	PRI	NULL	auto_increment
u_id	int(11)	NO	MUL	NULL	
property_id	int(11)	NO	MUL	NULL	
status	enum('pending','confirmed','cancelled')	YES		pending	
scheduled_date	datetime	NO		NULL	
created_at	timestamp	NO		CURRENT_TIMESTAMP	
amount_paid	decimal(10,2)	NO		0.00	
payment_status	varchar(20)	YES		pending	
payment_mode	varchar(50)	YES		NULL	
checkin_date	date	YES		NULL	
checkout_date	date	YES		NULL	

d. Transactions:

Field	Type	Null	Key	Default	Extra
transaction_id	int(11)	NO	PRI	<i>NULL</i>	auto_increment
u_id	int(11)	NO	MUL	<i>NULL</i>	
property_id	int(11)	NO	MUL	<i>NULL</i>	
amount	decimal(12,2)	NO		<i>NULL</i>	
payment_status	enum('pending','completed','failed')	YES		pending	
payment_method	enum('credit_card','bank_transfer','cash')	NO		<i>NULL</i>	
created_at	timestamp	NO		CURRENT_TIMESTAMP	

e. Messages:

Field	Type	Null	Key	Default	Extra
message_id	int(11)	NO	PRI	<i>NULL</i>	auto_increment
sender_id	int(11)	NO	MUL	<i>NULL</i>	
receiver_id	int(11)	NO	MUL	<i>NULL</i>	
message	text	NO		<i>NULL</i>	
sent_at	timestamp	NO		CURRENT_TIMESTAMP	

4) Relation among Tables:

